MIL-C-22931/12
10 February 1968
SUPERSEDING
MIL-C-22931/2A (in part)
27 March 1964

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY

SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC

.875 AND 1.005 INCHES, OUTSIDE DIAMETERS, 75 OHMS

The complete requirements for procuring cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: See table.

Dielectrics:
Cable core and insulating layer: See table.

Outer conductor: See table.

Jacket: See table.

Voltage withstand: See table.

Insulation defects, spark: See table.

Attenuation: See table.

Velocity: See table.

Capacitance: See table.

Impedance: 75 ±1.5 ohms.

Voltage standing wave ratio: See table.

Mandrel size for cold bend, bending and temperature cycling tests: See table.

Minimum bending radius: See table.

FSC 6145

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Requirement 1/	M22931/12 dash number		
	-001	-002	-003
Inner conductor	Copper wire or tube		Copper tube
Overall diameter	0.189(4.80)±.002(.05)		0.226(5.74) ±.001(.03)
Dielectrics			
Cable core	Polystyrene		Polyethylene
Insulating layer	Polystyrene	do.	None
Outer conductor	Aluminum tubing, smooth, seamless		Aluminum corrugated tube
Inside diameter (nominal)	0.758(19.25)		0.795(20.19)
Outside diameter	0.875(22.23)		1.005±.010 (25.53) (.25)
Jacket	Not applicable	Polyethylene	Polyethylene
Diameter		0.980(24.89) +.020(551) -.010(.25)	1.115±.030 (28.32) (.76)
Minimum wall thickness		0.049(1.24)	0.045(1.14)
Insulating defects, spark	Not applicable	Applicable	Applicable
Voltage withstand			
Volts rms at 60 Hz	3,200 nom		3,400
Volts dc	6,000		6,000
Attenuation (dB/100 ft max)			
30 MHz	0.25		0.26
400 MHz	1.00		0.98
3,000 MHz	1.64		1.60
Velocity (percent, nom)	89.7		90
Capacitance (pf/ft, nom)	14.7	do.	15.1
VSWR (max at 10-1,000 MHz)			
Initial	1.15		1.15
After temperature cycling	1.20		1.15
After bending	1.20		1.20
Mandrel size radius	17.5(444.5)		10(254.0)
Minimum bending radius	10(254.0)		10(254.0)

1/ Dimensions are in inches; millimeters are in parentheses. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

NOTES:

1. Cross reference information is as follows

Part number	RG number	Former classification	
		Class	Type
M22931/12-001	RG246/U	N	I
M22931/12-002	RG247/U		
M22931/12-003	None	None	None

2. Figures 1 and 2 are for information.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

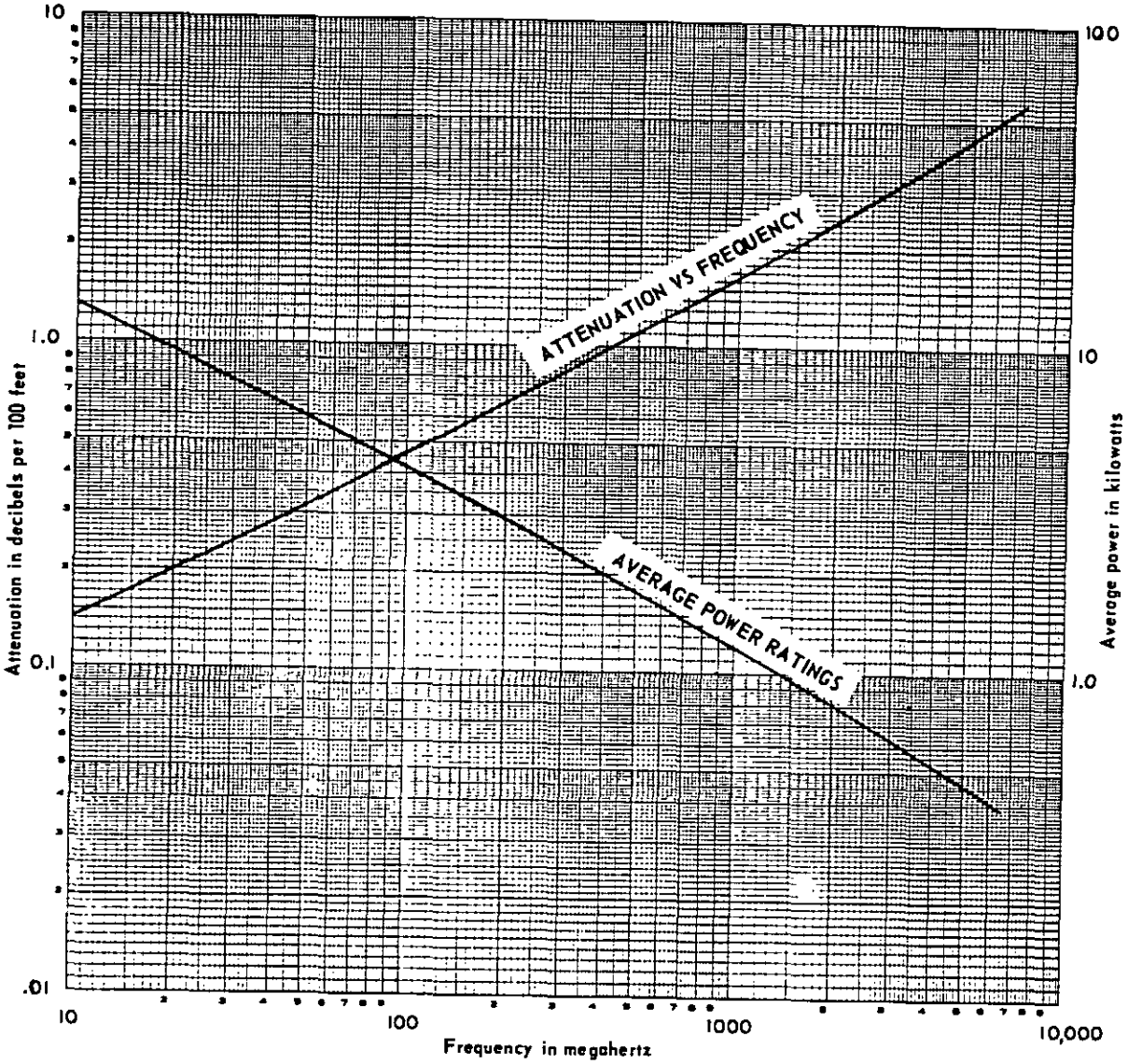
Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

(Project 6145-0305-3)

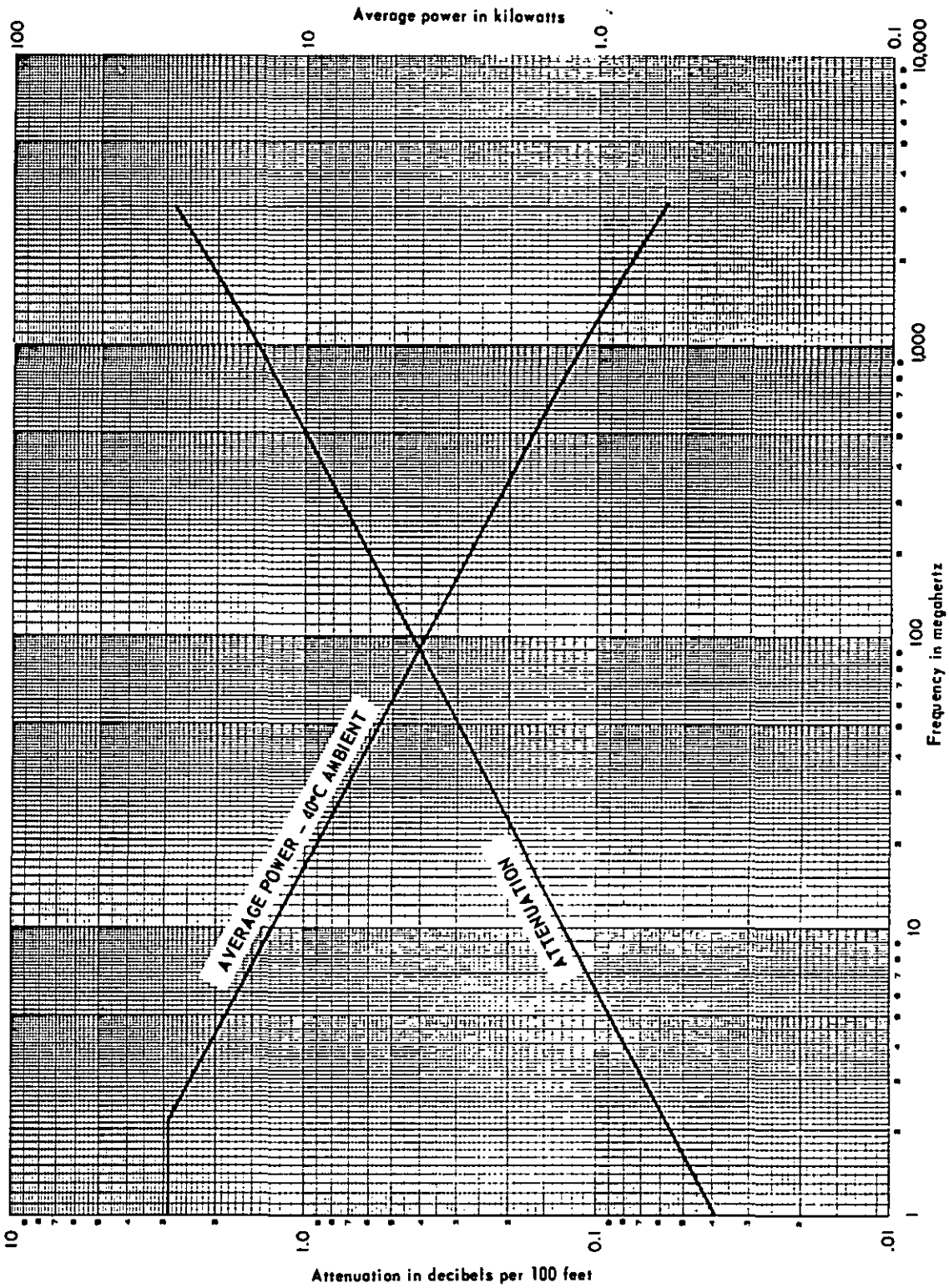
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for -001 and -002.

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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 2. Attenuation and power rating curves for –003.

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